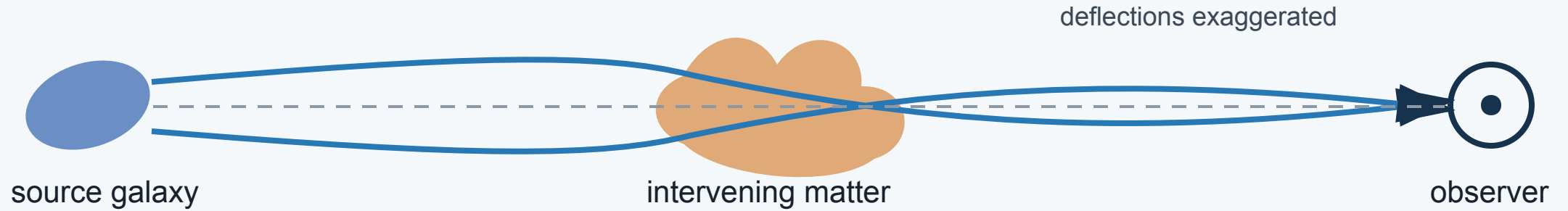
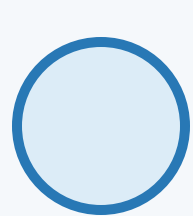


1. Light paths through intervening matter

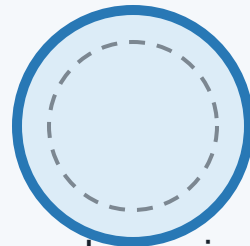
AI-generated — GPT-5.6 Sol



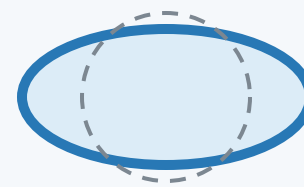
2. Convergence and shear change an image



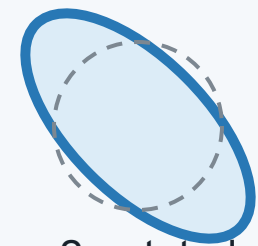
unlensed



kappa: larger image



gamma1: horizontal shear



gamma2: rotated shear

3. Spatial scale and wavenumber

Schematic density fluctuations over equal distance intervals

large scale $\rightarrow$ small k



long spatial wavelength

$$k = 2\pi / \text{spatial wavelength}$$

$$P_{\Delta}(k,z)$$

strength of density fluctuations at each scale and time

small scale $\rightarrow$ large k



short spatial wavelength